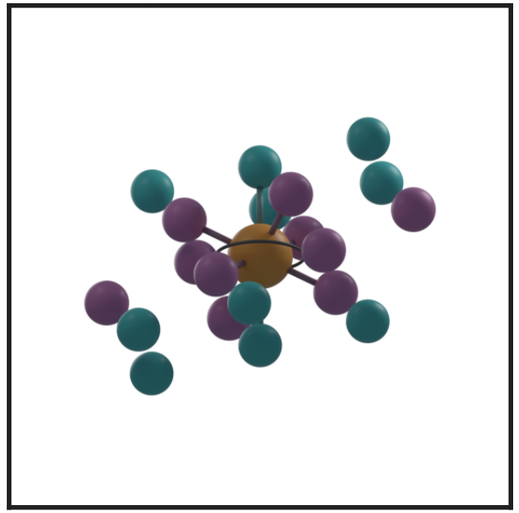
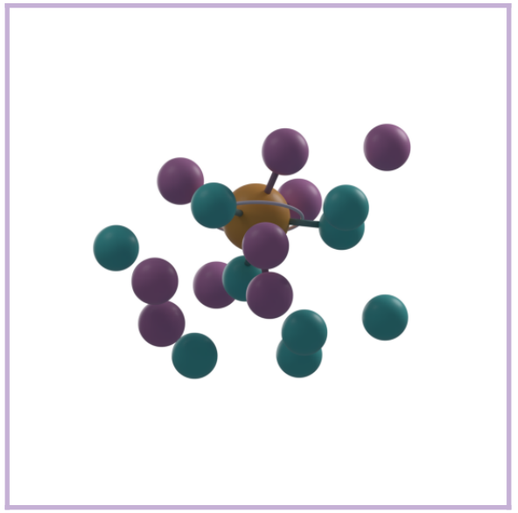
Coordination-sphere fingerprints of representative MigrationBench configurations

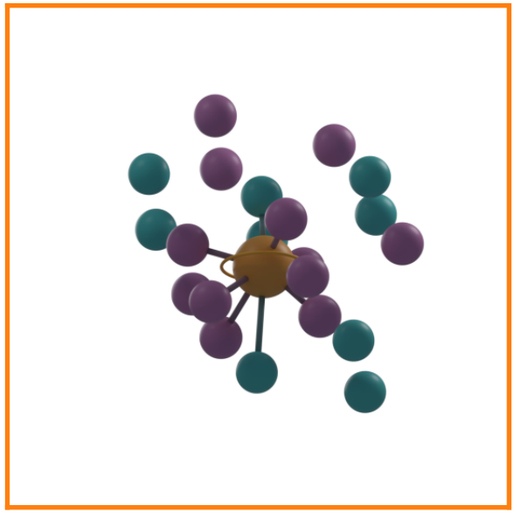
Perfect reference
Cr-host endpoint



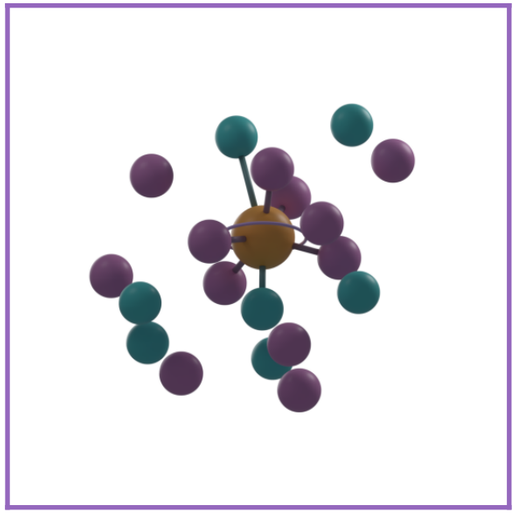
C10 · FT-600K self-NEB



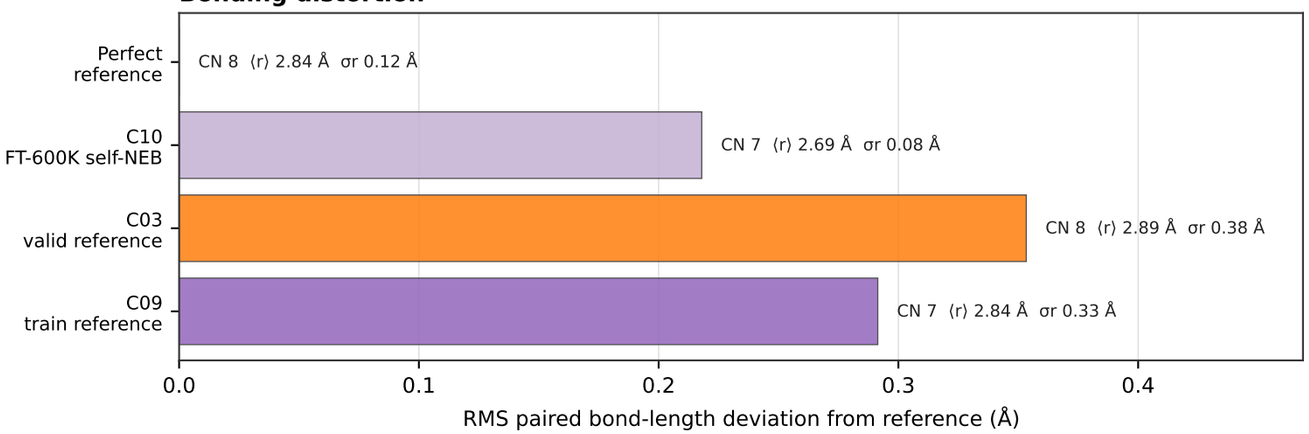
C03 · grouped valid reference



C09 · grouped train reference



Bonding distortion



Interpretation of the same coordination shell

- Perfect reference: 8 neighbours (2 Sb + 6 Te), narrow shell.
- C10: one fewer neighbour and a compressed, low-spread shell along the self-NEB frame.
- C03/C09: reference-set environments, but with larger shell spread or coordination loss.
- Cylinders are distance-qualified coordination guides; no bond order is inferred.

All panels use the same Cr-centred 3.75 Å shell. The reference is a Cr-containing host endpoint; an undoped pristine host has no Cr centre. RMSΔr is a structural descriptor, not an energy barrier or a model-ranking score.